

Mawazo mengi kuhusu AI na roboti si ya kweli

Zhihu • 2025-12-06

Chini ya kila swali kuhusu uhamishaji wa viwanda, mnyororo kamili wa viwanda, ukuaji wa viwanda n.k., kumejaa maoni kwamba otomatiki itafanya kazi ya kimwili kiwandani kuwa isiyo na maana. Maoni haya yanatumika kuthibitisha: 1) viwanda vya China havikuhitaji kuhamishwa; 2) faida ya nguvu kazi ya bei nafuu ya nchi zisizoendelea haitafanya kazi; 3) n.k.

Baada ya kutolewa kwa ChatGPT, uwezo wa AI umezua mawazo mengi. Wengi wanaona AI itawafukuza wafanyakazi wengi wa ofisi kazini. Vivyo hivyo, madereva wa kiotomatiki na roboti zitawafukuza wafanyakazi wa viwanda. Nguvu kazi ya binadamu itakuwa ya ziada; wengi watabaki bila kazi; hatimaye tatizo la usambazaji litasuluhishwa na mifumo kama mapato ya msingi.

Wengine hata wanaamini maneno ya Musk, Altman na wengine kwamba AGI — akili ya jumla ya bandia — inakaribia. AGI kama ubongo kwenye roboti inatoa roboti ya jumla. Pamoja na maonyesho ya makampuni ya roboti, wengi wanaamini “mustakabali umefika”.

Leo hatutajadili mageuzi ya ustaarabu wa kibepari baada ya roboti ya jumla. Hili ni kubwa, tutajadili baadaye. Makala hii inajadili mipaka ya wimbi hili la maendeleo ya AI na roboti.

Mifano kubwa ya lugha (LLM)

Kwa kuwa mimi mwenyewe ninatumia mifano mikubwa kazini kwa otomatiki, katika matumizi halisi niligundua sifa zifuatazo:

- 1) Inafanya vizuri sana kwa tokeni nyingi za kuingiza na tokeni chache za kutoa: uchaguzi wa API, uchimbaji wa vigezo, kupanga kazi kwa muktadha mwingi.
- 2) Tokeni nyingi za kuingiza na kutoa, lakini bila ubunifu: kwa mfano, tafsiri.
- 3) Tokeni chache za kuingiza, nyingi za kutoa, bila hitaji la mantiki: kwa mfano, ubunifu wa fasihi na sanaa.

Kazi hizi tayari kwa kiasi kikubwa zinaweza kuchukua nafasi ya kazi ya binadamu. Lakini kuunganisha uwezo huu wa LLM katika mchakato wa kazi kunahitaji ubinafsishaji

mwingi — iwe prompts maalum au mafunzo ya ziada ya mfano. Na mifano baada ya mwanzoni mwa mwaka huu, inaonekana, haikuboreka sana. Hata kwa mafunzo na uboreshaji wote, matokeo ya kibiashara yenye usahihi wa 90% ni mazuri tayari.

Kwa mfano, tulifikiria kutengeneza kodi moja kwa moja kutoka maelezo ya mahitaji, hata tukachagua mfano uliofunzwa hasa kwa upangaji. Lakini kwa kweli, ikiwa maelezo ya muktadha hayatoshi, kodi inayotengenezwa daima inakosa kitu; kupata kodi inayokubalika inahitaji mazungumzo ya mara kwa mara. Na “kodi inayokubalika” hapa inamaanisha visa vya msingi vinapitia tu, si kwamba hakuna hitilafu. Wanaotumia wanajua: kodi ya AI imejaa hitilafu. Bidhaa kwenye “ubongo wa LLM” zinazoweza kuchukua nafasi ya mtu — haiwezekani.

LLM inaweza kuchukua nafasi ya nini? Huduma za wateja, michakato ya ofisi. Kwa sababu kazi nyingi ni kuchimba vigezo kutoka lugha ya asili ya wateja na kujaza mfumo wa IT; kazi kama hizo ni rahisi kubadilisha na LLM. Lakini tu. Kufanya mifano mikubwa kuwafukuza wafanyakazi wengi wa ofisi kazini — haiwezekani.

Ripoti za hivi karibuni zinazothibitisha mapovu ya AI zinaonyesha tatizo hili.

Kwa mfano, ripoti ya MIT: 95% ya makampuni yanayotumia AI ya kuzalisha inashindwa — yaani, ingawa makampuni mengi yanafuata wimbi la AI, kwa kweli haifanyi kazi. Kuna ripoti nyingine kadhaa, hitimisho sawa.

Search

FORTUNE

Home

News

Fortune 500

Tech

Finance

Leadership

Lifestyle

Rankings

Multimedia

NEWSLETTERS • CFO DAILY

MIT report: 95% of generative AI pilots at companies are failing



By Sheryl Estrada

Senior Writer And Author Of CFO Daily

Add us on





August 18, 2025, 6:54 AM ET



知乎 @最遥远的路

GETTY IMAGES

Mtu atasema: je uwezo wa mifano mikubwa hautakua zaidi? Kuendelea kuongeza GPU, vigezo, kufundisha mifano mikubwa zaidi. Kwa kweli, njia ya “nguvu kubwa” ya scaling law imefikia kikomo.

Napendekeza makala ya mhandisi wa Google:

Fundamental Limits of Large Language Model Scaling: Analysis of Mathematical Impossibility Theorems and Resource Constraints

<https://medium.com/hackrlife/fundamental-limits-of-large-language-model-scaling-analysis-of-mathematical-impossibility-theorems-53ece210e23e>

Makala imekusanya makala 25 za hivi karibuni, ikithibitisha — kutoka kutowezekana kwa kudhibiti ukumbusho wa uwongo, kukauka kwa data, mipaka ya data ya sintetiki, Performance Plateaus and Diminishing Returns (plato la utendaji na kupungua kwa faida katika scaling), mipaka ya alignment — kwamba LLM-AI kwenye usanifu wa sasa iko hivyo, na kuendelea kuiboresha kupitia scaling law ni vigumu.

Kwa mfano, makala inayofuata inazungumzia kutoendana kwa vigezo na faida. Kati ya ukubwa wa mfano na uboreshaji wa utendaji kuna uhusiano wa logarithmic: kila kuongezeka mara mbili kwa ukubwa kunatoa faida ndogo.

6.1 Emerging Patterns in Scaling Limitations

Our analysis has revealed several key patterns that characterize the current state of LLM scaling:

- The relationship between model size and performance improvements follows a logarithmic curve, with each doubling of parameters yielding progressively smaller gains. This pattern is particularly evident in benchmark results where GPT-4's 76.60% accuracy on math tasks and Claude 3.5 Sonnet's 90.20% accuracy in code evaluations (Vellum Research Team, 2024) represent significant but increasingly difficult-to-surpass achievements.
- The resource requirements for crossing higher emergence thresholds (θ_c) grow exponentially, while the corresponding performance improvements grow logarithmically. This misalignment between cost and benefit suggests that continued scaling alone cannot be the primary path forward.
- Data quality and diversity have emerged as potentially more critical constraints than raw computational power. The variance term $V(P, D)$ in our bias-variance decomposition increasingly dominates performance limitations, indicating that data innovation may be more crucial than model size scaling.

Has LLM Reached the Scaling Ceiling Yet? Unified Insights into LLM Regularities and Constraints

<https://arxiv.org/pdf/2412.16443>

Makala inayofuata inathibitisha: ukumbusho wa uwongo hauwezi kuepukika. Hii si usanifu wa LLM, bali matokeo ya mkusanyiko wa habari, kwa hiyo hauepukiki.

9 Summary

We have established that no large language model can simultaneously achieve perfect hallucination (or imagination) control. We showed that four essential properties—truthful knowledge representation, semantic information conservation, complete revelation of relevant knowledge, and optimal response generation—are mutually incompatible. This impossibility emerges from the mathematical structure of information aggregation itself, not from limitations in data, compute, or architecture.

We introduced a rigorous mathematical framework and grounded in it the general analysis of inference processes and their LLM implementations. At the heart of our mathematical framework lies the semantic information measure μ_C and the emergence operator $\mathcal{E}_C$. The earlier one is a context-dependent metric capturing how knowledge reduces uncertainty within computational bounds. The later one formalizes how reasoning makes latent knowledge explicit rather than creating it anew. Finally, we model LLM inference as an auction of ideas—a idealized marketplace where neural components compete by bidding with their partial knowledge, each trying to influence the final response.

The impossibility manifests through three complementary proofs, each illuminating different aspects of the fundamental constraint. Through game theory, we apply the Green-Laffont theorem to show the impossibility when knowledge components are independent. Through probability theory, we leverage Savage's proper scoring rules to extend the result to general trained or fine-tuned probability distributions. And through direct analysis of transformer architectures, we demonstrate how log-sum-exp convexity in attention mechanisms creates measurable violations of information conservation. The last proof exploits Jensen inequality for convex mappings to define the precise mathematical quantity that measures how aggregation of ideas (auction bids) produces overconfident responses that the constituent components cannot justify with revealed knowledge.

These mathematical insights we have made yield some interesting philosophical implications. Hallucination and creativity (two sides of the same coin we call imagination) emerge as mathematically identical phenomena, distinguished only by our normative judgments. The impossibility result explores the outer limits of reason, identified by Gödel's incompleteness in mathematics, Heisenberg's uncertainty in physics, and Arrow's impossibility in social choice. Also, our framework seems to provide mathematical foundations and experimental settings for models of artificial consciousness. That may open interesting avenues for future research.

On the Fundamental Impossibility of Hallucination Control in Large Language Models
<https://arxiv.org/pdf/2506.06382>

Makala nyingine, msomaji anayevutiwa anaweza kuzisoma mwenyewe.

Kwa bahati, mwandishi wa GPT Ilya Sutskever hivi karibuni pia anasema: enzi ya scaling imeisha; ni wakati wa kurudi maabara kuchunguza usanifu mpya. Kwa maneno mengine, kipindi hiki cha mlipuko wa kibiashara wa AI kimeisha; tena maabara, hadi mruko ujao wa kiteknolojia utakapofungua mzunguko mpya wa kibiashara. Anaweza kuwa na motisha zake, lakini katika sekta wamekuwa wakisema hivi. Ni kwamba yeye ndiye mwandishi wa Scaling Law, kwa hiyo "hukumu" yake ina mamlaka.

Ilya Sutskever Declares the **Scaling Era Dead**. His \$3 Billion Bet Says Research Will Win.

Ilya Sutskever helped prove that scale works in AI. Now he says it doesn't—and his \$3 billion company is betting the future belongs to researchers with ideas, not labs with the biggest GPU clusters. The industry isn't ready for this argument.



MARCUS SCHULER

NOVEMBER 25 2025 • 10:20 AM (UTC-08:00 • PST) • 10 MIN READ

SHARE ▼



Roboti

Wazo moto zaidi kuhusu roboti sasa: roboti za umbo la binadamu zinaweza kuzoea miundombinu iliyojengwa kuzunguka mfanyakazi wa binadamu, na kuchukua nafasi ya kazi ya binadamu kwa upole. Mara tu roboti za umbo la binadamu zitakapopata "wakati wake wa ChatGPT", mabilioni ya wafanyakazi wa kimwili duniani wanaweza kubadilishwa — hii itakuwa soko kubwa.

Mimi sifanyi kazi kwenye roboti, sina uzoefu. Lakini daima nimetilia shaka maneno kwamba roboti zitawafanya wafanyakazi wa mstari wa kusanyiko kuwa wasiohitajika.

Robot factories could threaten jobs of millions of garment workers



📷 Workers on a production line in Ho Chi Minh City, Vietnam. Nine million people, mostly young women, are dependent on garment and footwear jobs in south-east Asia. Photograph: Richard Vogel/AP

As Adidas announces a new robot shoe factory, an ILO report says up to 90% of workers in south-east Asia could face unemployment due to automation

The jobs of nearly 90% of garment and footwear workers in Cambodia and Vietnam are at risk from automated assembly lines - or “sewbots” - according to a new [report](#) from the International Labour Organisation (ILO).

There are 9 million people, mostly young women, dependent upon jobs in textiles, garments, and footwear within the Association of Southeast Asian Nations (Asean) economic area, which includes [Cambodia](#), Indonesia, Thailand and Malaysia. These are the workers the ILO identifies as most

知乎 @最最遥远的路

Miaka 9 imepita — je roboti zimebadilisha wafanyakazi wa viwanda vya nguo vya Bangladesh na Vietnam? Hapana. Bangladesh pekee ina wafanyakazi milioni 5 wa nguo.

Over 50.17 lakh people working in garment sector in Bangladesh: state minister

Staff Correspondent 23 June, 2024, 21:39 [Best clothing retailers](#)

f Share

X Tweet

✉ Email

Share

<

Print



Representational image. [Travel guide book](#)

There are over 50.17 lakh workers working in garment factories in the country and 55.57 per cent of them are women, state minister for labour and employment Nazrul Islam Chowdhury told Jatiya Sangsad on Sunday.

知乎 @最遥远的路

Hivi karibuni niliona makala ya profesa wa robotiki wa MIT Rodney Brooks. Yeye ni mwanachama wa Chuo cha Sayansi cha Australia, mkurugenzi wa maabara ya sayansi ya kompyuta na AI ya MIT, mmoja wa waanzilishi wa iRobot; amefanya kazi miongo kadhaa kwenye robotiki — hakika mtaalamu.

Katika makala "Why Today's Humanoids Won't Learn Dexterity" anataja sababu mbili muhimu:

1. Ujifunzaji wa roboti za kisasa unategemea kuona, lakini ustadi wa mkono wa binadamu hautegemei kuona tu, bali mguso. Mfumo uliofunzwa bila data ya mguso hautafikia ustadi wa "mkono". Ugumu wa njia hii: data ya mafunzo ya mguso ni vigumu kupata kuliko ya kuona. Utafiti wa sasa wa maabara ni wa awali sana; hakuna mazungumzo ya kibiashara.

2. Uwezo wa roboti za sasa ni mdogo; haziwezi kuchukua nafasi ya mtu mara moja, kwa hiyo lazima zifanye kazi na mtu. Lakini kufanya kazi kwa roboti za umbo la binadamu na mtu katika nafasi moja, karibu sana, ni hatari sana. Ushauri wake: usikaribie roboti za umbo la binadamu la ukubwa kamili...

My advice to people is to not come closer than 3 meters to a full size walking robot. And the walking robot companies know this too. Even in their videos you will not see people close to a locomoting humanoid robot unless there is a big table between them, and even then the humanoids only shuffle around a little bit.

Kwa utabiri wake wa kiufundi Predictions Scorecard, 2025 January 01:

1) Kabla ya 2030, mkono wa roboti wenye ustadi hauwezekani; anatumai 2040.

Dexterous robot hands generally available.

NET 2030
BY 2040 (I hope!)

Despite some impressive lab demonstrations we have not actually seen any improvement in widely deployed robotic hands or end effectors in the last 40 years.

2) Roboti ya kiwango cha paka au mbwa haitatokea kabla ya 2048.

A robot that seems as intelligent, as attentive, and as faithful, as a dog.

NET 2048

This is so much harder than most people imagine it to be--many think we are already there; I say we are not at all there.

3) Roboti inayoelewa binadamu kama mtoto wa miaka 6 — haitaonekana katika maisha yake (kabla ya 2050).

A robot that has
any real idea
about its own
existence, or the
existence of
humans in the
way that a six
year old
understands
humans.

知乎 @最最遥远的路

Utabiri huu unarekebisha kila mara kwa hali ya sekta, lakini jambo dhahiri: roboti hizi za umbo la binadamu zinazolenga kibiashara hazina uwezo wa kuchukua nafasi ya mtu kabisa; hata kama zitaweza kuingia kwenye maeneo ya kazi ya binadamu kusaidia — swali kubwa.

Madereva wa kiotomatiki

Madereva wa kiotomatiki ni matumizi yaliyokomaa zaidi ya AI; wengi wanaona kazi ya dereva itatoweka baada ya miaka michache. Lakini wataalamu wa sekta hawaoni hivyo.

Tathmini ya Rodney Brooks: uuzaji kamili wa madereva wa kiotomatiki hauwezekani kabla ya 2045. Magari ya sasa ya “kibiashara” ya kujisukuma ama yana dereva wa dhamana, ama yanaendeshwa na mtu nyuma — sio madereva wa kiotomatiki halisi.

The majority of US
cities have the
majority of their
downtown under
such rules.

知乎 @最最遥远的路

Hype ya madereva wa kiotomatiki imekuwa mawimbi kadhaa; huu ni mwisho.

Some of the AI Hype Cycles that I Have Seen

- Pre-me:
 - Neural networks (vol 1), 1943, McCulloch & Pitts
 - Turing, 1950 computers fast enough for AGI
 - Neural networks (vol 2) 1950s, Minsky & Rosenblatt
 - McCarthy, 1956 computers fast enough for AGI
- Michie, 1959, reinforcement learning (1)
- Minsky, Michie, Nilsson, 1960: search
- It is all first order logic & resolution theorem proving
- Neural networks (vol 3) 1968 Minsky again
- MacHack (search, chess 1)
- Fuzzy logic, STRIPS, knowledge-based systems (revolution in medicine 1)
- Neural networks (vol 4) back propagation
- Primal sketch, 1979, Marr
- Self driving cars 1, 1987, Dickmanns
- 1980s reinforcement learning (2, & 3)
- SOAR, qualitative reasoning, support vector machines
- Self driving cars 2, 1997, Kanade et al
- Deep Blue (search, chess 2)
- Self driving cars 3, 2007, Thrun
- Bayesian inference
- Watson (Jeopardy & revolution in medicine 2)
- Deep learning, neural networks (vol 5)
- Self driving cars 4, 2010+, commercial
- Alpha Go, reinforcement learning (4)
- Generative AI, neural networks (vol 6)

Watengenezaji wa magari wanatangaza madereva kamili kwa miaka, kama makampuni ya magari ya China yanavyodai kufikia L3, lakini kwa kweli L3 inafikiwa tu katika matukio fulani; baada ya masharti mengi, sio L3 tena.

FORECASTS: http://www.driverless-future.com/?page_id=384 March 27, 2017

- NVIDIA to introduce level-4 enabling system by 2018 (2017)
- NuTonomy to provide self-driving taxi services in Singapore by 2018, expand to 10 cities around world by 2020 (2016)
- Delphi and MobilEye to provide off-the-shelf self-driving system by 2019 (2016)
- Ford CEO announces fully autonomous vehicles for mobility services by 2021 (2016) ←
- Volkswagen expects first self driving cars on the market by 2019 (2016)
- GM: Autonomous cars could be deployed by 2020 or sooner (2016) ←
- BMW to launch autonomous iNext in 2021 (2016) ←
- Ford's head of product development: autonomous vehicle on the market by 2020 (2016) ←
- Baidu's Chief Scientist expects large number of self-driving cars on the road by 2019 (2016)
- First autonomous Toyota to be available in 2020 (2015) ←
- Elon Musk now expects first fully autonomous Tesla by 2018, approved by 2021 (2015)
- US Sec Trans: Driverless cars will be in use all over the world by 2025 (2015)
- Uber fleet to be driverless by 2030 (2015) ←
- Ford CEO expects fully autonomous cars by 2020 (2015) ←
- Next generation Audi A8 capable of fully autonomous driving in 2017 (2014)
- Jaguar and Land-Rover to provide fully autonomous cars by 2024 says Director of Research and Technology (2014)
- Fully autonomous vehicles could be ready by 2025, predicts Daimler chairman (2014) ←
- Nissan to provide fully autonomous vehicles by 2020 (2013) ←
- Truly autonomous cars to populate roads by 2028-2032 estimates insurance think tank executives (2013)
- Continental to make fully autonomous driving a reality by 2025 (2012)

Hitimisho

Katika hali ya Scaling Law kufikia kikomo, mipaka ya matumizi mengine ya AI imeamuliwa. Roboti na madereva wa kiotomatiki, kwa ustadi wa kiwango cha binadamu, vinahitaji si tu utambuzi, bali uwezo wa kufikiri. Kufikiri ni kwa kanuni au kwa mtandao wa neva. Kwa kanuni — daima kuna matukio ya pembeni yasiyozingatwa; kwa mtandao wa neva — jinsi ya kutatua kutokuwa sahihi. Katika uwanja wa kazi ya kidijitali ya ofisi, pengine kuna nafasi ya kujaribu. Katika uwanja wa kazi ya kimwili, ambapo usalama wa maisha unahusika, suala lenyewe ni jinsi ya kueneza. Lakini bila kueneza, data za mafunzo ya ziada zitapatikana wapi? Kuku na yai — lipi la kwanza?

Maendeleo ya AI, roboti, madereva wa kiotomatiki kwa binadamu yanapaswa kuwa mazuri; lakini ikiwa teknolojia hizi zitatumika tu kwa malengo ya mtaji, zinaweza kuleta maafa kwa binadamu. Kwa bahati nzuri, teknolojia haziendelei kwa nguvu kama Musk na wengine wanavyojivunia; binadamu bado ana nafasi ya kuendeleza na kutumia AI katika mfumo bora wa kijamii.

Makala hii ilichapishwa awali kwenye Zhihu (kwa Kichina): <https://zhuanlan.zhihu.com/p/1980678884421822141> Mwandishi: 最遥远的路 • Ukurasa wa Zhihu: <https://www.zhihu.com/people/zui-zui-yao-yuan-de-lu-32/posts>

本文原发表于知乎专栏/回答 (中文): <https://zhuanlan.zhihu.com/p/1980678884421822141>; 作者: 最遥远的路: <https://www.zhihu.com/people/zui-zui-yao-yuan-de-lu-32/posts>